1. Difference between Local Storage Options

shared_preferences

- Stores **small data** in **key-value pairs**
- Data types: String, int, double, bool, List<String>
- Example: login status, user settings
- Very easy to use

SQLite

- Works like a **relational database (tables + rows)**
- Stores **large structured data**
- Supports **SQL queries**
- Example: user data, products, attendance records

Hive

- **NoSQL database** (key-value, but faster than SQLite)
- No need for SQL queries
- Works well with **Flutter (pure Dart)**
- Example: offline apps, caching, complex objects

Quick Difference Table

Feature	shared_preferences	SQLite	Hive
Data Type	Small key-value	Structured	Key-value
Speed	Fast	Medium	Very Fast
Complexity	Easy	Hard	Medium
Use Case	Settings	Large apps	Offline apps

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2. CRUD Operations (Create, Read, Update, Delete)

Meaning

- **C → Create** (Insert data)
- **R → Read** (Fetch data)

- **U → Update** (Modify data)
- **D → Delete** (Remove data)

In SQLite (Example concept)

- **Create**

INSERT INTO users VALUES(1, 'Jeet');

- **Read**

SELECT * FROM users;

- **Update**

UPDATE users SET name='Raj' WHERE id=1;

- **Delete**

DELETE FROM users WHERE id=1;

In Hive (Concept)

- **Create**

box.put('name', 'Jeet');

- **Read**

box.get('name');

- **Update**

box.put('name', 'Raj');

- **Delete**

box.delete('name');

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3. Advantages & Use Cases of shared_preferences (Only Theory)

Advantages

- Very **simple and easy**
- No database setup required
- Fast for small data
- Good for storing **permanent app settings**

Use Cases

- User login status (isLoggedIn)
- Theme (dark/light mode)
- Language selection
- App preferences
- Small configuration data